Haozhuang Chi

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01 PROFILE

Passionate and motivated PhD, interested in AI, LLM & its application (World Model VLA), Autonomous driving, and Robotics.

02 EDUCATION

Jan 2023 — Jan 2027
Singapore

Nanyang Technological University

Ph.D

Doctor of Philosophy | School of Mechanical and Aerospace Engineering (MAE)

Supervised by Prof. Lv Chen @ Automan

Research focus

- Autonomous driving
- Embodied AI
- End-to-End Model (Large language model)
- World Model + VLA (Vision Language Action Model)

Funding(Full Scholarship PhD): Singapore International Graduate Award (SINGA) from Agency for Science, Technology & Research (A*STAR) and Nanyang Technological University

Apr 2020 — Nov 2022
Munich

Technical University of Munich

Master of Science (M.Sc.)

Robotics, Cognition, Intelligence

Supervised by Prof. Daniel Rueckert @MCML - Munich Center for Machine Learning

Daniel Rückert is Alexander von Humboldt Professor also a Professor at Imperial College London. He served as Head of the Department of Computing at Imperial College. He received Germany's highest research honor, the prestigious Gottfried Wilhelm Leibniz Prize for his groundbreaking work in AI-assisted medical imaging.

Sept 2016 — Mar 2020
Stuttgart

University of Stuttgart

Bachelor

Mechanical Engineering, Bachelor

Sept 2014 — Sept 2015
Beijing

China University of Mining and Technology(Peking)

Bachelor

Mechanical Engineering, Bachelor

03 SCHOLARSHIP & AWARD

Jan 2023 — Dec 2026
Singapore

Singapore International Graduate Award (SINGA) at Agency for Science, Technology & Research (A*STAR) &Nanyang Technological University

The Singapore International Graduate Award (SINGA) is a collaboration between the Agency for Science, Technology & Research (A*STAR), the Nanyang Technological University (NTU), the National University of Singapore (NUS), the Singapore University of Technology and Design (SUTD) and the Singapore Management University (SMU).

Jan 2023 — Dec 2026
Singapore

NTU Research Scholarship (RSS) at Nanyang Technological University - NTU Singapore

The NTU Research Scholarship is awarded to outstanding graduate students for research.

04 ACTIVITIES

Aug 2025 — Present
Singapore

PhD Research Collaborator at Continental AG

As a PhD Research Collaborator in the Scalable Perception Project , a joint initiative between Continental AG and Nanyang Technological University (NTU), I contribute research efforts in developing AI-based situational awareness systems through driver monitoring and environmental perception fusion.

Project Implementation: 2027 CES, 2028 IAA Mobility.

Jan 2023 — Present
Singapore

PhD student at Nanyang Technological University

Main Research focus

- Autonomous driving
- Embodied AI
- End-to-End Model (Large language model)
- World Model + VLA (Vision Language Action Model)

Large Language Models (LLMs) for Autonomous Systems: By integrating LLMs with autonomous driving technologies, we are setting new benchmarks for vehicle intelligence, enabling them to comprehend and navigate complex environments with unprecedented accuracy and efficiency. This approach not only advances the field of autonomous vehicles but also serves as a cornerstone for developing intelligent agents within the industrial Metaverse.

Apr 2022 — Oct 2022

Munich

Master Thesis at Lab for AI in Medicine, Technical University of Munich

Image Denoising Networks: Overview of the State-Of-The-Arts in the 2020s

Recently a great number of frequency networks with CNNs are used in this field to improve the reconstruction quality. More recently, the advent of Transformers evolutes the computer vision field. Its inherent attributes like a larger receptive field and lower inductive bias facilitate the image denoising tasks.

A comparison with SOTA methods (NAFNet/Restormer/SwinIR/Uformer/DnCNN, etc.) on public datasets like urban100 or SIDD will be conducted in the end.

Sept 2021 — Mar 2022

Ingolstadt

Intern at Audi

Development for augmented reality head-up displays.

I work closely with colleagues in the head-up display development department and have contacts to other development departments as well as quality, procurement, sales, production and other group brands, such as CARIAD.

I support the employees of our team in the development of future display concepts and technologies.

I support us in operational activities such as technology investigations, acquisition of concept data, vehicle tests, supplier control or the creation of committee documents.

In addition, l receive challenging project tasks that I work on and supervise independently as part of the internship.

Apr 2022 — Oct 2022

Munich

Practical Training at Informatik 7 - Theoretical Foundations of Artificial Intelligence, Technical University of Munich

Practical - Analysis of new phenomena in machine/deep learning

Deep neural networks produce state-of-the-art results on a wide range of machine learning problems. While deep learning still remains elusive to rigorous theoretical analysis, its phenomenal performance has shaken the mathematical foundations of machine learning—contradicting many conventional beliefs of classical learning theory and the fundamental understanding of how algorithms can successfully learn patterns. However, the past few years have seen an exciting combination of mathematical and empirical research, uncovering some of the mysteries of modern machine/deep learning and development of formal theories to explain them, such as:

- Generalization
- Over-parameterization
- Kernel behavior
- Robustness
- Unsupervised learning

Familiarize with empirical practices used to explain surprising phenomena in deep learning. In particular, reproduce empirical findings of recent papers from top ML conferences (Neurips, ICML, ICLR and AISTATS) and empirically extend these observations to more complex problems/models.

Oct 2021 — Feb 2022

Munich

Practical Training at Informatics 6 - Chair of Robotics, Artificial Intelligence and Real-time Systems, TUM Department of Informatics, Technical University of Munich

Cloud-Based Machine Learning in Robotics

Applying state-of-the-art deep learning methods to robotics remains challenging. Generating the required amount of data for training on physical robots is costly and usually takes too long to be practically feasible. This is why simulation environments are becoming increasingly important in machine learning. They not only reduce costs and setup times but also enable arbitrary acceleration of the learning process through massively parallel deployment in the cloud. However, many state-of-the-art simulation environments only support very simple robot systems are not tailored to the specific requirements in robotics.
The goal is to set up virtual environments in a cloud-based robot simulation environment and to train machine learning models in the cloud. Participants will leverage modern deplyoment methods such as containers to deploy detailed simulation models in the cloud and train simulated robot to perform tasks based on simulated sensor input.

Solving a Reaching Task with Reinforcement Learning and Spiking Neural Networks in the NRP

Solving a reaching task with a robot manipulator and Spiking Neural Networks (SNNs) in the Neurorobotics Platform. Every episode, a new target point in 3D will be generated randomly, and the goal is to direct the manipulator's joint angles such that the end effector reaches that point. Observations are going to be ground truth data from the simulation such as the target pose and the current joint angles and velocities. Review the state-of-the-art reinforcement learning algorithms and select a suitable one to solve this task, keeping in mind that rewards are going to be sparse. Adapt the RL algorithm to SNNs, and think about the encoding and decoding of inputs and outputs.

Oct 2021 — Feb 2022

Munich

Advanced Seminar at Chair of Application and Middleware Systems, TUM Department of Computer Science

Large-Scale Graph Processing and Graph Partitioning

Graph Neural Network - Architectures

In this seminar we study several large-scale (distributed) graph processing systems for static and dynamic graphs and graph neural networks. Furthermore, we study streaming and in-memory graph partitioners.
Acquiring an in-depth understanding about:
-(Distributed) large-scale Graph Processing
-(Distributed) neural networks
-Graph Partitioning
-Graph Clustering and Community Detection
-Static and Dynamic Graphs
-Hypergraphs
-In-memory and streaming Partitioners

An understanding about different Graph Neural Network architectures like for example:

Graph Attention Networks (GATs), Graph Convolutional Networks (GCN), GraphSAGE, Relational Graph Convolutional Network (RGCN), etc.

Apr 2021 — Oct 2021

Munich

Practical training at Chair of Robotics, Artificial Intelligence and Real-time Systems, Technical University of Munich

Development of Biologically-inspired Methods for Autonomous Driving

To build Biologically-inspired algorithms for Autonomous Driving

Learn about autonomous driving

- Sensors including Camera, LiDAR, DVS

- Mapping, Localization, Object detection, Navigation…

- DNN, Reinforcement Learning

Look at state-of-art biologically-inspired methods

- Place cells, Head direction cells, Grid Cells

- Spiking neural networks

Build algorithm

Localization and Mapping with Brain-inspired Neuron Models

- Utilizing place cell, head direction cell model to solve the SLAM problem.

Use ROS with C++ and Gazebo to simulate the RatSLAM.

Jul 2020 — Oct 2020

Munich

Seminar at Computer Vision Group, Technical University of Munich

Recent Advances in 3D Computer Vision

Joint Texture and Geometry Optimization for RGB-D Reconstruction

Apr 2019 — Oct 2019

Stuttgart

Bachelor thesis at Institute for Control Engineering of Machine Tools and Manufacturing Units (ISW), University of Stuttgart

Expert system for determining production parameters

Nov 2018 — Mar 2019

Stuttgart

Project work at Institute of Internal Combustion Engines and Automotive Engineering (IVK)(FA), University of Stuttgart

Application of deep learning in the Powertrain development

05 SKILLS

Python

LLM

ROS

Linux

MatLab

Microsoft Office

OpenStudio

Raspberry Pi

C++

Docker

Pytorch

Tensorflow

EnergyPlus

Java

Doxygen

SketchUp

Nginx

AutoCAD

GT-Power

06 LANGUAGES

English	●●●●●	Cantonese	●●●●○
German	●●●●●	Japanese	●○○○○
Mandarin	●●●●●		

07 PUBLICATIONS

2025

VLM-DM: Visual Language Models for Multitask Domain Adaptation in Driver Monitoring, IEEE IV

Authors: Haozhuang Chi, Haohan Yang, Lie Yang, and Chen Lv *

2025

Applications of Large Language Models and Multimodal Large Models in Autonomous Driving: A Comprehensive Review, Drones

Authors: Jing Li, Jingyuan Li, Guo Yang, Lie Yang *, Haozhuang Chi, Lichao